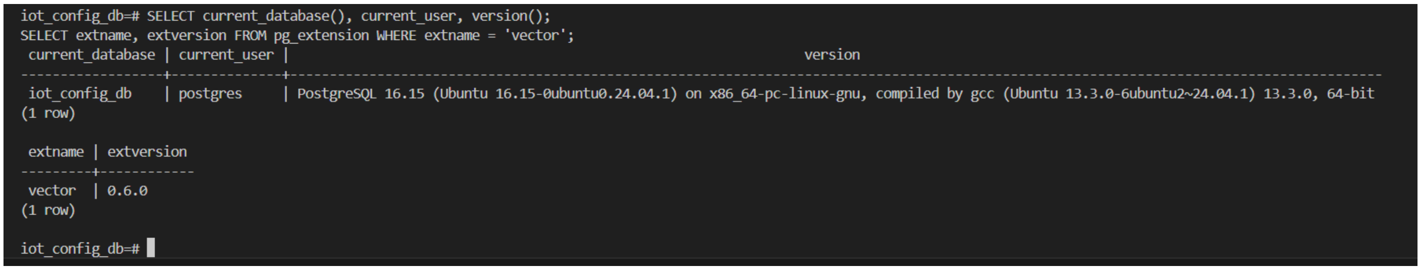
DBMS Experiential Learning — Level 3 Mini-Project

Live DDL & Database Query Execution Evidence Report
Project Title: IoT Device Configuration & Compatibility Database
Team Number: 9 | **Database:** PostgreSQL 16.15 + pgvector (v0.6.0)
Students: 1. Hitha Shree Suresh (PES2UG24CS196) | 2. Hemant Ram Nelapati (PES2UG24CS192)

1. PostgreSQL 16 Connection & pgvector Extension Setup

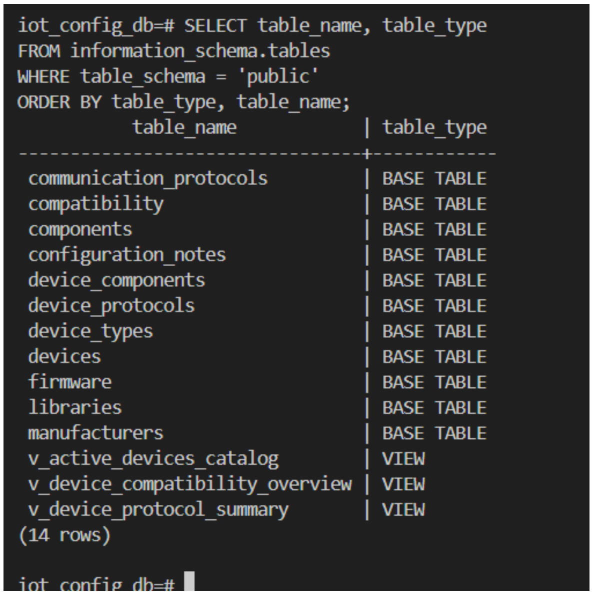
Verification of connection to database iot_config_db and active vector (v0.6.0) extension.



```
iot_config_db=# SELECT current_database(), current_user, version();
SELECT extname, extversion FROM pg_extension WHERE extname = 'vector';
 current_database | current_user |                                                              version
------------------+--------------+------------------------------------------------------------------------------------------------------------------------------------------
 iot_config_db    | postgres     | PostgreSQL 16.15 (Ubuntu 16.15-0ubuntu0.24.04.1) on x86_64-pc-linux-gnu, compiled by gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0, 64-bit
(1 row)

 extname | extversion
---------+------------
 vector  | 0.6.0
(1 row)

iot_config_db=#
```

2. Schema Introspection (11 Base Tables + 3 Views)

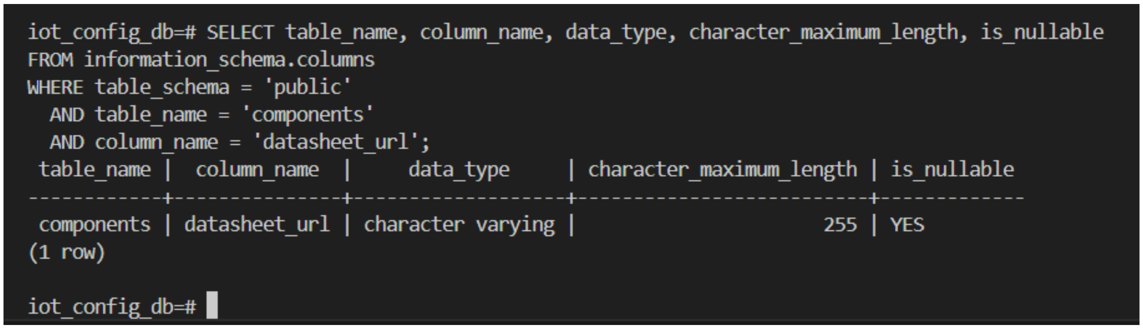
PostgreSQL information_schema.tables catalog verifying all 11 relational tables and 3 views.



```
iot_config_db=# SELECT table_name, table_type
FROM information_schema.tables
WHERE table_schema = 'public'
ORDER BY table_type, table_name;
           table_name           | table_type
--------------------------------+------------
 communication_protocols        | BASE TABLE
 compatibility                  | BASE TABLE
 components                     | BASE TABLE
 configuration_notes            | BASE TABLE
 device_components              | BASE TABLE
 device_protocols               | BASE TABLE
 device_types                   | BASE TABLE
 devices                        | BASE TABLE
 firmware                       | BASE TABLE
 libraries                      | BASE TABLE
 manufacturers                  | BASE TABLE
 v_active_devices_catalog       | VIEW
 v_device_compatibility_overview | VIEW
 v_device_protocol_summary      | VIEW
(14 rows)

iot_config_db=#
```

3. Schema Evolution via ALTER TABLE

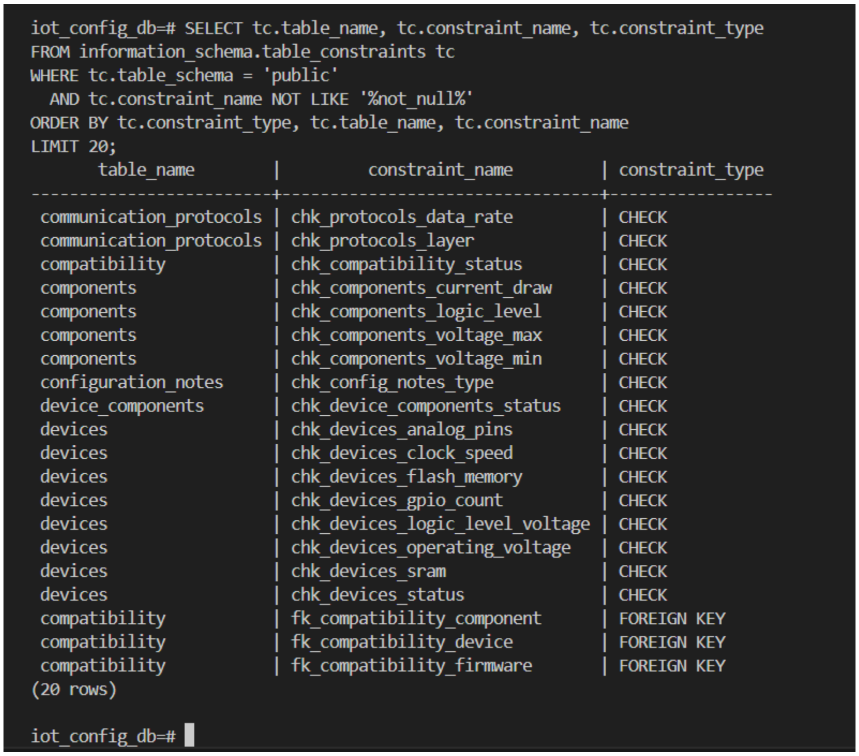
Demonstration of ALTER TABLE components ADD COLUMN datasheet_url VARCHAR(255); in catalog.



```
iot_config_db=# SELECT table_name, column_name, data_type, character_maximum_length, is_nullable
FROM information_schema.columns
WHERE table_schema = 'public'
  AND table_name = 'components'
  AND column_name = 'datasheet_url';
 table_name |  column_name  |     data_type     | character_maximum_length | is_nullable
------------+---------------+-------------------+--------------------------+-------------
 components | datasheet_url | character varying |                      255 | YES
(1 row)

iot_config_db=#
```

4. Constraint Enforcement (PK, FK, UNIQUE, CHECK Constraints)

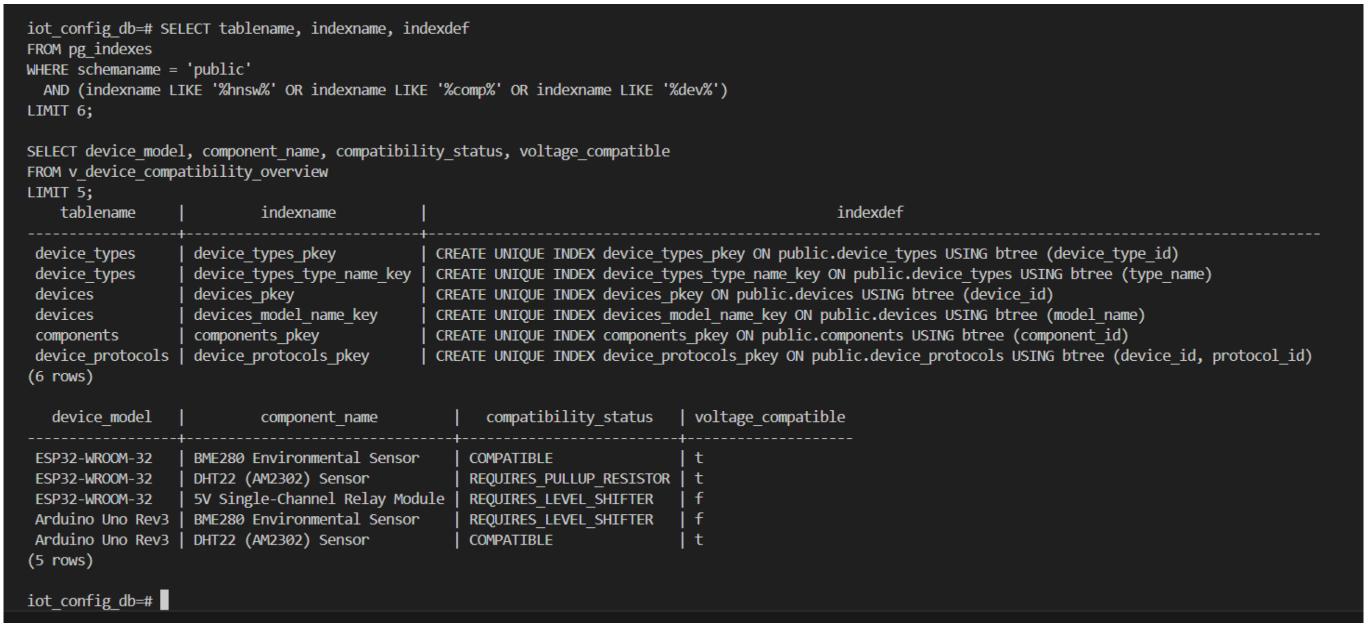
Introspection of domain CHECK rules, foreign keys, primary keys, and unique constraints in table_constraints.



```
iot_config_db=# SELECT tc.table_name, tc.constraint_name, tc.constraint_type
FROM information_schema.table_constraints tc
WHERE tc.table_schema = 'public'
  AND tc.constraint_name NOT LIKE '%not_null%'
ORDER BY tc.constraint_type, tc.table_name, tc.constraint_name
LIMIT 20;
       table_name        |         constraint_name         | constraint_type
-------------------------+---------------------------------+-----------------
 communication_protocols | chk_protocols_data_rate         | CHECK
 communication_protocols | chk_protocols_layer             | CHECK
 compatibility           | chk_compatibility_status        | CHECK
 components              | chk_components_current_draw     | CHECK
 components              | chk_components_logic_level      | CHECK
 components              | chk_components_voltage_max      | CHECK
 components              | chk_components_voltage_min      | CHECK
 configuration_notes     | chk_config_notes_type           | CHECK
 device_components       | chk_device_components_status    | CHECK
 devices                 | chk_devices_analog_pins         | CHECK
 devices                 | chk_devices_clock_speed         | CHECK
 devices                 | chk_devices_flash_memory        | CHECK
 devices                 | chk_devices_gpio_count          | CHECK
 devices                 | chk_devices_logic_level_voltage | CHECK
 devices                 | chk_devices_operating_voltage   | CHECK
 devices                 | chk_devices_sram                | CHECK
 devices                 | chk_devices_status              | CHECK
 compatibility           | fk_compatibility_component      | FOREIGN KEY
 compatibility           | fk_compatibility_device         | FOREIGN KEY
 compatibility           | fk_compatibility_firmware       | FOREIGN KEY
(20 rows)

iot_config_db=# 
```

5. Advanced Feature 1 & 2: Analytical Views & HNSW Vector Index

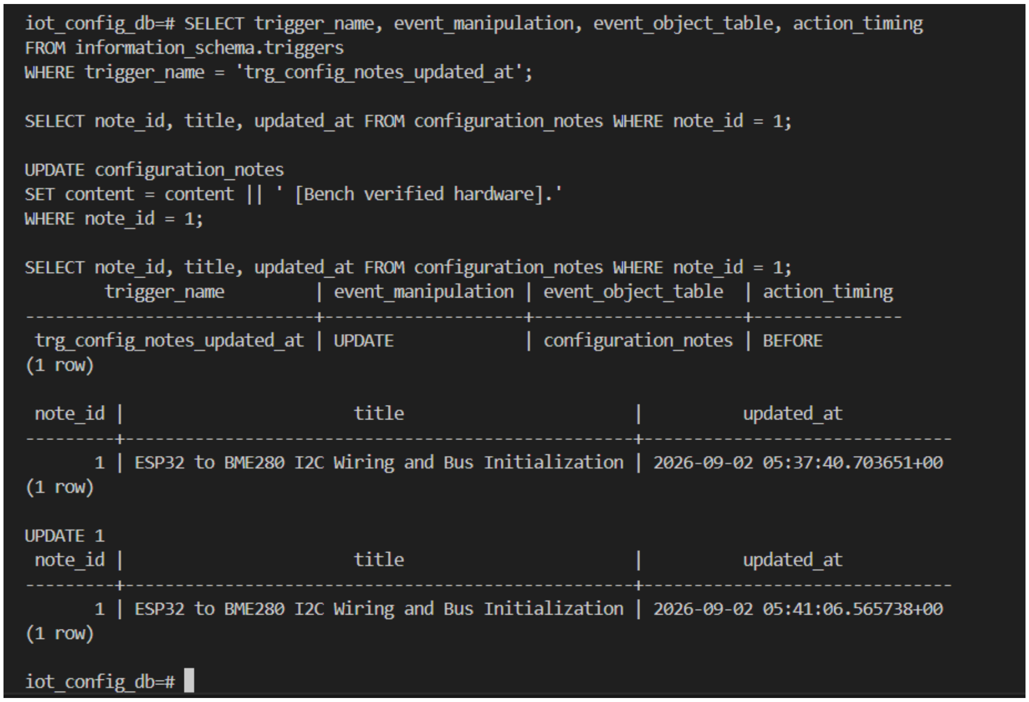
Query execution against v_device_compatibility_overview and HNSW vector index catalog.



```
iot_config_db=# SELECT tablename, indexname, indexdef
FROM pg_indexes
WHERE schemaname = 'public'
  AND (indexname LIKE '%hnsw%' OR indexname LIKE '%comp%' OR indexname LIKE '%dev%')
LIMIT 6;

SELECT device_model, component_name, compatibility_status, voltage_compatible
FROM v_device_compatibility_overview
LIMIT 5;
    tablename     |         indexname          |                                                  indexdef
------------------+----------------------------+-----------------------------------------------------------------------------------------------------------
 device_types     | device_types_pkey          | CREATE UNIQUE INDEX device_types_pkey ON public.device_types USING btree (device_type_id)
 device_types     | device_types_type_name_key | CREATE UNIQUE INDEX device_types_type_name_key ON public.device_types USING btree (type_name)
 devices          | devices_pkey               | CREATE UNIQUE INDEX devices_pkey ON public.devices USING btree (device_id)
 devices          | devices_model_name_key     | CREATE UNIQUE INDEX devices_model_name_key ON public.devices USING btree (model_name)
 components       | components_pkey            | CREATE UNIQUE INDEX components_pkey ON public.components USING btree (component_id)
 device_protocols | device_protocols_pkey      | CREATE UNIQUE INDEX device_protocols_pkey ON public.device_protocols USING btree (device_id, protocol_id)
(6 rows)

   device_model   |         component_name         |   compatibility_status   | voltage_compatible
------------------+--------------------------------+--------------------------+--------------------
 ESP32-WROOM-32   | BME280 Environmental Sensor    | COMPATIBLE               | t
 ESP32-WROOM-32   | DHT22 (AM2302) Sensor          | REQUIRES_PULLUP_RESISTOR | t
 ESP32-WROOM-32   | 5V Single-Channel Relay Module | REQUIRES_LEVEL_SHIFTER   | f
 Arduino Uno Rev3 | BME280 Environmental Sensor    | REQUIRES_LEVEL_SHIFTER   | f
 Arduino Uno Rev3 | DHT22 (AM2302) Sensor          | COMPATIBLE               | t
(5 rows)

iot_config_db=# 
```

6. Advanced Feature 3: Database Trigger & Stored Function

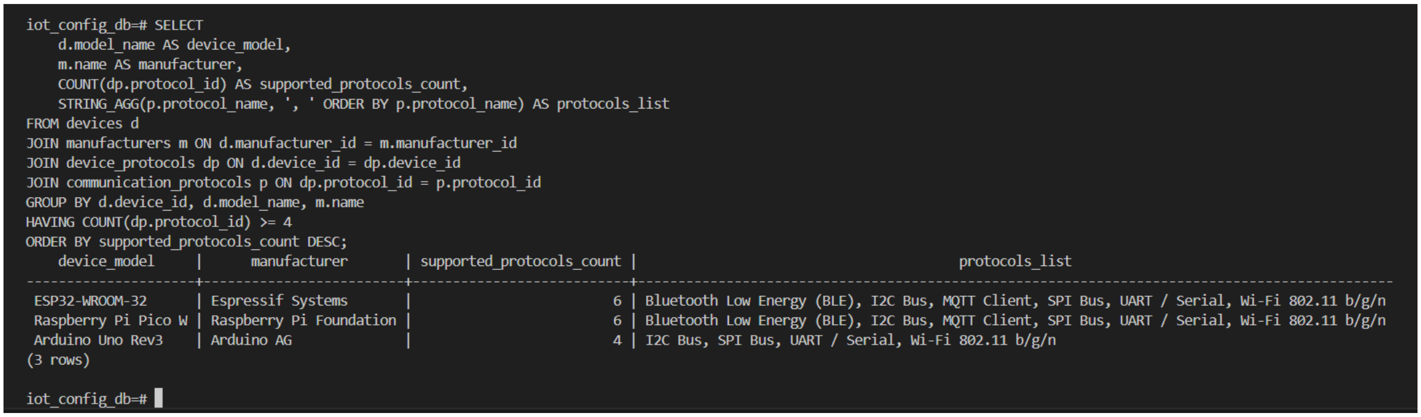
Verification of trg_config_notes_updated_at automatically updating the updated_at timestamp on ROW UPDATE.



```
iot_config_db=# SELECT trigger_name, event_manipulation, event_object_table, action_timing
FROM information_schema.triggers
WHERE trigger_name = 'trg_config_notes_updated_at';

SELECT note_id, title, updated_at FROM configuration_notes WHERE note_id = 1;

UPDATE configuration_notes
SET content = content || ' [Bench verified hardware].'
WHERE note_id = 1;

SELECT note_id, title, updated_at FROM configuration_notes WHERE note_id = 1;
        trigger_name         | event_manipulation | event_object_table  | action_timing
-----------------------------+--------------------+---------------------+---------------
 trg_config_notes_updated_at | UPDATE             | configuration_notes | BEFORE
(1 row)

 note_id |                       title                       |          updated_at
---------+---------------------------------------------------+-------------------------------
       1 | ESP32 to BME280 I2C Wiring and Bus Initialization | 2026-09-02 05:37:40.703651+00
(1 row)

UPDATE 1
 note_id |                       title                       |          updated_at
---------+---------------------------------------------------+-------------------------------
       1 | ESP32 to BME280 I2C Wiring and Bus Initialization | 2026-09-02 05:41:06.565738+00
(1 row)

iot_config_db=# █
```

7. DML Operations: Aggregation with GROUP BY and HAVING Clause

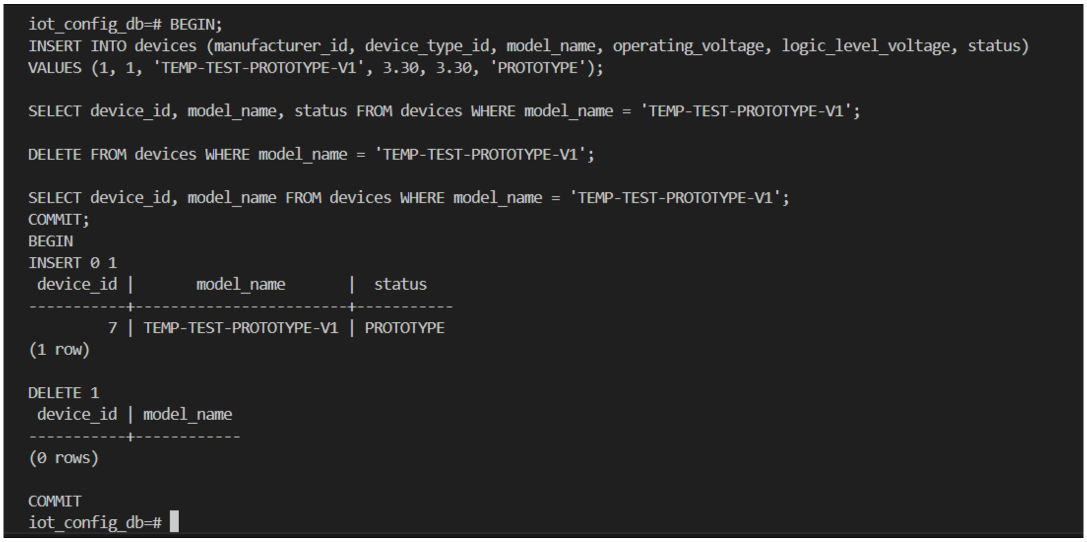
Analytical query filtering IoT boards supporting at least 4 communication protocols (HAVING COUNT(dp.protocol_id) >= 4).



```
iot_config_db=# SELECT
    d.model_name AS device_model,
    m.name AS manufacturer,
    COUNT(dp.protocol_id) AS supported_protocols_count,
    STRING_AGG(p.protocol_name, ', ' ORDER BY p.protocol_name) AS protocols_list
FROM devices d
JOIN manufacturers m ON d.manufacturer_id = m.manufacturer_id
JOIN device_protocols dp ON d.device_id = dp.device_id
JOIN communication_protocols p ON dp.protocol_id = p.protocol_id
GROUP BY d.device_id, d.model_name, m.name
HAVING COUNT(dp.protocol_id) >= 4
ORDER BY supported_protocols_count DESC;
    device_model     |      manufacturer       | supported_protocols_count |                                       protocols_list
---------------------+-------------------------+---------------------------+----------------------------------------------------------------------------------------------
 ESP32-WROOM-32      | Espressif Systems       |                         6 | Bluetooth Low Energy (BLE), I2C Bus, MQTT Client, SPI Bus, UART / Serial, Wi-Fi 802.11 b/g/n
 Raspberry Pi Pico W | Raspberry Pi Foundation |                         6 | Bluetooth Low Energy (BLE), I2C Bus, MQTT Client, SPI Bus, UART / Serial, Wi-Fi 802.11 b/g/n
 Arduino Uno Rev3    | Arduino AG              |                         4 | I2C Bus, SPI Bus, UART / Serial, Wi-Fi 802.11 b/g/n
(3 rows)

iot_config_db=# █
```

8. DML Operations: Transaction-Safe DELETE Statement

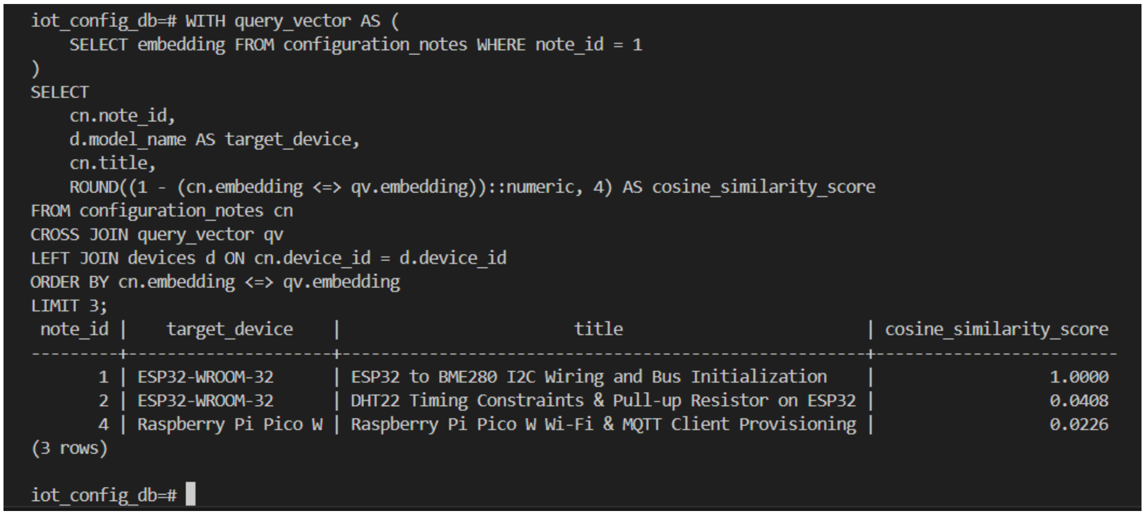
Demonstration of safe row insertion, existence verification, deletion, and rollback/commit within an ACID transaction.



```
iot_config_db=# BEGIN;
INSERT INTO devices (manufacturer_id, device_type_id, model_name, operating_voltage, logic_level_voltage, status)
VALUES (1, 1, 'TEMP-TEST-PROTOTYPE-V1', 3.30, 3.30, 'PROTOTYPE');

SELECT device_id, model_name, status FROM devices WHERE model_name = 'TEMP-TEST-PROTOTYPE-V1';

DELETE FROM devices WHERE model_name = 'TEMP-TEST-PROTOTYPE-V1';

SELECT device_id, model_name FROM devices WHERE model_name = 'TEMP-TEST-PROTOTYPE-V1';
COMMIT;
BEGIN
INSERT 0 1
 device_id |       model_name       |  status
-----------+------------------------+-----------
         7 | TEMP-TEST-PROTOTYPE-V1 | PROTOTYPE
(1 row)

DELETE 1
 device_id | model_name
-----------+------------
(0 rows)

COMMIT
iot_config_db=# ▌
```

9. Advanced Feature 4: pgvector Cosine Similarity Semantic Search

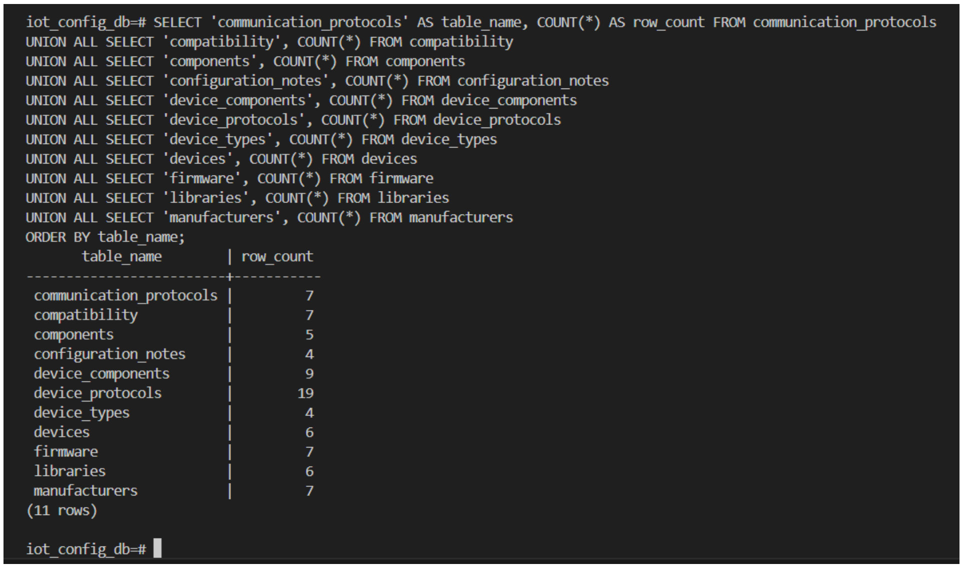
Semantic query matching technical documentation notes using cosine distance operator (<=>).



```
iot_config_db=# WITH query_vector AS (
    SELECT embedding FROM configuration_notes WHERE note_id = 1
)
SELECT
    cn.note_id,
    d.model_name AS target_device,
    cn.title,
    ROUND((1 - (cn.embedding <=> qv.embedding))::numeric, 4) AS cosine_similarity_score
FROM configuration_notes cn
CROSS JOIN query_vector qv
LEFT JOIN devices d ON cn.device_id = d.device_id
ORDER BY cn.embedding <=> qv.embedding
LIMIT 3;
 note_id |    target_device    |                        title                         | cosine_similarity_score
---------+---------------------+------------------------------------------------------+-------------------------
       1 | ESP32-WROOM-32      | ESP32 to BME280 I2C Wiring and Bus Initialization    |                  1.0000
       2 | ESP32-WROOM-32      | DHT22 Timing Constraints & Pull-up Resistor on ESP32 |                  0.0408
       4 | Raspberry Pi Pico W | Raspberry Pi Pico W Wi-Fi & MQTT Client Provisioning |                  0.0226
(3 rows)

iot_config_db=# ▌
```

10. Database Population Verification (81 Seed Records Total)

Row count verification across all 11 normalized tables confirming exactly 81 seed records.



```
iot_config_db=# SELECT 'communication_protocols' AS table_name, COUNT(*) AS row_count FROM communication_protocols
UNION ALL SELECT 'compatibility', COUNT(*) FROM compatibility
UNION ALL SELECT 'components', COUNT(*) FROM components
UNION ALL SELECT 'configuration_notes', COUNT(*) FROM configuration_notes
UNION ALL SELECT 'device_components', COUNT(*) FROM device_components
UNION ALL SELECT 'device_protocols', COUNT(*) FROM device_protocols
UNION ALL SELECT 'device_types', COUNT(*) FROM device_types
UNION ALL SELECT 'devices', COUNT(*) FROM devices
UNION ALL SELECT 'firmware', COUNT(*) FROM firmware
UNION ALL SELECT 'libraries', COUNT(*) FROM libraries
UNION ALL SELECT 'manufacturers', COUNT(*) FROM manufacturers
ORDER BY table_name;
       table_name        | row_count
-------------------------+-----------
 communication_protocols |         7
 compatibility           |         7
 components              |         5
 configuration_notes     |         4
 device_components       |         9
 device_protocols        |        19
 device_types            |         4
 devices                 |         6
 firmware                |         7
 libraries               |         6
 manufacturers           |         7
(11 rows)

iot_config_db=# ▌
```